Soumya Swayamjyoti

Computational Materials Scientist
Machine Learning Researcher
Visiting Researcher at RWTH Aachen University

Doctor of Science (Dr. sc.) – ETH Zurich Germany / Switzerland / India

Google Scholar Profile

Research Profile
Materials scientist with expertise in atomistic simulations, metallic glasses, computational mechanics, machine learning in materials science, finite element methods, scientific computing, AI-driven materials discovery, and interdisciplinary research connecting materials science, photonics, and machine learning.

Academic Degrees

Doctor of Science (Dr. sc.) in Materials Science ETH Zurich, Switzerland, 2012–2016

Doctoral research conducted jointly at:

- Laboratory of Metal Physics and Technology, ETH Zurich
- Condensed Matter Theory Group, Paul Scherrer Institute

Thesis: Atomistic simulations to study metallic glasses: a microscopic investigation of local structural excitations

Doctoral thesis link

Master of Science in Computational Mechanics of Materials and Structures University of Stuttgart, Germany, 2009–2012

One semester of studies at UPC Barcelona, Spain.
Thesis: Finite element implementation of orbital-free density functional theory

Master thesis link

Bachelor of Technology in Civil Engineering NIT Karnataka, Surathkal, India, 2004–2008

Thesis: Analysis of plates by Navier method

Current Position

Visiting Researcher Institute of General Mechanics, RWTH Aachen University, Germany

Research activities in:

- Computational mechanics

- Materials modelling
- Machine learning for engineering systems
- Scientific computing
- Multi-scale simulations
- Data-driven modelling

Previous Academic and Professional Experience

- **Scientific Researcher**, School of Minerals, Metallurgical and Materials Engineering (SMMME), IIT Bhubaneswar, India, 2019–2025.
- **Postdoctoral Research Associate**, School of Engineering, Brown University, USA, 2017.
- **Research Engineer**, Siemens Technology Center, Bangalore, India, 2018.
- **Software Engineer and AI Developer**, Hyscaler Consulting Group, Bhubaneswar, India, 2018.
- **Doctoral Researcher**, ETH Zurich and Paul Scherrer Institute, Switzerland, 2012–2016.
- **Graduate Student Assistant**, University of Stuttgart, Germany, 2010–2012.
- **Graduate Student Assistant**, CIMNE Barcelona, Spain, 2009.
- **Software Engineer**, Intergraph Process, Power and Marine, Hyderabad, India, 2009.
- **Software Engineer**, subsidiary of Exeter Group, Bangalore, India, 2008.

Research Areas

- Materials Science
- Metallic Glasses
- Atomistic Simulations
- Molecular Dynamics
- Computational Mechanics
- Finite Element Methods
- Scientific Computing
- Machine Learning for Materials
- Deep Learning
- AI for Scientific Discovery
- Data-driven Modelling
- Numerical Optimization
- Photonics Collaborations
- High-performance Computing

Publications

Google Scholar Profile

Complete and updated list of publications, citations, and research metrics:
https://scholar.google.com/citations?user=w6ZvEUwAAAAJ

Selected Publications

1. Sakthi Priya Amirtharaj, Zujun Xu, **Donna Strickland** , Borun Chowdhury, Sagnik Acharya, Priyam Samantray, Anil Prabhakar, Kisor Kumar Sahu, Franz Bamer, and S. Swayamjyoti,

   "Numerically optimized FROG results for the study of red-shifted spectra in multi-frequency Raman generation,"

   arXiv preprint, 2026.

   Special distinction: collaboration with 2018 Nobel Prize winner Prof. Donna Strickland.

2. S. Swayamjyoti, J. F. Löffler, and P. M. Derlet,

   "Local structural excitations in model glasses,"

   Physical Review B, 89, 224201, 2014.

3. S. Swayamjyoti, J. F. Löffler, and P. M. Derlet,

   "Local structural excitations in model glass systems under applied load,"

   Physical Review B, 93, 144202, 2016.

4. R. Kishore, S. Swayamjyoti, S. Das, A.K. Gogineni, Z. Nussinov, D. Solenov, K.K. Sahu,

   "Visual Machine Learning: Insight through Eigenvectors, Chladni patterns and community detection in 2D particulate structures,"

   accepted in *International Journal for Multiscale Computational Engineering*.

5. S. Das, H. Pegu, K.K. Sahu, A.K. Nayak, S. Ramakrishna, D. Datta, S. Swayamjyoti,

   "Machine Learning in Materials Modeling – Fundamentals and the Opportunities in 2D Materials,"

   Elsevier, 2020.

   Role: Principal investigator and corresponding author.

6. Abhilash M Menon, Tanmayee R Kopparthi, Pravan Omprakash, Harikesh Verma, Ayan Haldar, S. Swayamjyoti, Kisor K Sahu, Carol Featherston,

   "Deep Learning-Based Optimization of Piezoelectric Vibration Energy Harvesters,"

   AIAA SCITECH Forum, 2022.

7. A.K. Gogineni, S. Swayamjyoti, D. Sahoo, K.K. Sahu, R. Kishore,

   "Multi-Class classification of vulnerabilities in Smart Contracts using AWD-LSTM,"

   IOP SciNotes, 2020.

Research Funding and Grants

- **Swiss National Science Foundation Research Grant** (2012–2016)

  Funding for doctoral research at ETH Zurich and Paul Scherrer Institute.

  Approximate support: 200,000+ CHF.

- **Erasmus Mundus Scholarship** (2009–2012)

  European Commission funded scholarship for master's studies.

  Approximate support: 42,000 EUR.

- **Research Funding** by Ministry of Human Resource Development and Ministry of Steel, India, for research projects at IIT Bhubaneswar.

Awards and Honours

- Postdoctoral Fellowship, Brown University, USA.
- Swiss National Science Foundation funded doctoral researcher, ETH Zurich.
- Erasmus Mundus Full Scholarship recipient.
- Selected for Entrepreneur First funded residency program.
- Research collaboration with Nobel Prize winning group.

Teaching and Mentoring

- Mentored PhD, master's, and undergraduate students.
- Co-instructor for materials science and engineering courses.
- Co-instructor for "Introduction to Machine Learning for Engineers and Materials Scientists" for 100+ undergraduate students.
- Guided research in: machine learning, materials modelling, molecular simulations, and computational mechanics.

Conference Presentations and Workshops

1. "Atomic-scale structural excitations in metallic glasses under zero and applied load," IIT Madras, India, 2019.
2. "Microscopic structural excitations in model metallic glass systems," Junior Euromat Conference, Lausanne, Switzerland, 2014.
3. CECAM Workshop, ETH Zurich, Switzerland, 2013.
4. CCMX Summer School on Multi-scale Modelling of Materials, Switzerland, 2012.
5. DUNE Workshop, Heidelberg, Germany, 2011.

Invited Talks

1. IIT Kharagpur, India, 2019.
2. Paul Scherrer Institute, Switzerland, 2012.
3. Max Planck Institute for Intelligent Systems, Germany, 2011.

Technical Skills

- Python
- C/C++
- MATLAB
- Fortran
- Machine Learning
- Deep Learning

- Molecular Dynamics
- Finite Elements
- Scientific Computing
- Linux
- T-SQL
- HTML/CSS/JavaScript
- ASP.NET
- HPC Computing

Languages

- English – Fluent (C2)
- Hindi – Native
- Oriya – Native
- German – A2 Level

Leadership and International Academic Engagement

- Initiated international collaboration with Nobel Prize winning research group.
- Led interdisciplinary collaborations involving researchers from: Germany, Switzerland, India, Singapore, Canada, and USA.
- Participated in Erasmus Mundus Association conferences.
- Organized venue coordination for EMA General Assembly, Barcelona, Spain.
- Featured in EMANATE alumni success stories.

Entrepreneurial and Industry Experience

- Selected for Entrepreneur First residency program.
- Industry experience in: software engineering, AI systems, scientific computing, machine learning, and enterprise technologies.

Career Breaks

None.